

**Applying Exponent Laws**Question Bank 3.05 **ANSWER KEY**

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*Show your work. Circle your final answer.***For questions 1 through 4, rewrite each expression as an equivalent expression with a positive exponent.**

1.  $\left(\frac{3}{4}\right)^{-3}$

$\left(\frac{4}{3}\right)^3$

2.  $\left(-\frac{7}{8}\right)^{-2}$

$\left(-\frac{8}{7}\right)^2$

*The sign stays with the fraction.*

3.  $\left(-\frac{4}{5}\right)^{-4}$

$\left(\frac{5}{-4}\right)^4$

4.  $\left(\frac{22}{7}\right)^{-7}$

$\left(\frac{7}{22}\right)^7$

**Matching.**

5. Match each expression on the left with the equivalent expression on the right by writing the letter.

a.  $\left(-\frac{11}{10}\right)^{-3}$

A.  $\left(-\frac{11}{10}\right)^3$

B.  $\left(\frac{10}{11}\right)^3$

C.  $\left(\frac{11}{10}\right)^3$

D.  $\left(-\frac{10}{11}\right)^3$

**D**

b.  $\left(\frac{11}{10}\right)^{-3}$

**B**

c.  $\left(-\frac{10}{11}\right)^{-3}$

**A**

d.  $\left(\frac{10}{11}\right)^{-3}$

**C**

For questions 6 through 9, generate an equivalent expression with one base and one positive exponent. Show the laws you use.

6.  $\left(-\frac{14}{17}\right)^{-2} \cdot \left(-\frac{14}{17}\right)^{-4}$

$$\left(-\frac{14}{17}\right)^{-6} = \left(-\frac{17}{14}\right)^6$$

Product of powers, then flip.

7.  $\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$

$$\left(\frac{5}{8}\right)^2$$

$-2 - (-4) = 2$ ; already positive.

8.  $\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$

$$\left(\frac{2}{3}\right)^9$$

$(5 - 2) \cdot 3$ .

9.  $\left(\frac{2.5^5}{2.5^3}\right)^{-1}$

$$2.5^{-2} = \left(\frac{5}{2}\right)^{-2} = \left(\frac{2}{5}\right)^2$$

Rewrite the decimal as a fraction before flipping.

**Equivalent or not.**

10. Consider the expression  $\frac{\left(\frac{1}{6}\right)^3}{6^{-5}} \cdot 6^4$ . For each expression below, decide whether it is equivalent, and justify with a law of exponents.

a.  $\frac{6^{-3}}{6^{-5}} \cdot 6^4$

**Yes** —  $\left(\frac{1}{6}\right)^3 = 6^{-3}$ , so this is the same expression; value  $6^6$ .

b.  $6^{-8} \cdot 6^4$

**No** — the quotient gives  $6^2$ , not  $6^{-8}$ ; this is  $6^{-4}$ .

c.  $\left(\frac{1}{6}\right)^{-2} \cdot 6^4$

**Yes** —  $\left(\frac{1}{6}\right)^{-2} = 6^2$ , and  $6^2 \cdot 6^4 = 6^6$ .

d.  $6^6$

**Yes** — it is the value of the expression written as one power.

11. Which is the value of  $\left(-\frac{2}{3}\right)^{-3}$  ?

☒ A.  $-\frac{27}{8}$

☐ B.  $\frac{27}{8}$

☐ C.  $-\frac{8}{27}$

☐ D.  $\frac{8}{27}$

**A**

$$(-3/2)^3 = -27/8.$$

12. Find the value of  $n$  that makes each statement true.

a.  $\left(\frac{3}{5}\right)^n = \frac{25}{9}$

**$n = -2$**

$$(5/3)^2 = 25/9.$$

b.  $\left(\frac{1}{2}\right)^n = 8$

**$n = -3$**